

SkyTech Technology Stack Use

Version V1.2 · 2026-10-10 · matches repository release R4.2 and architecture diagram V1.7 · maintained by SkyTech_Manager

This document explains, in plain words, which tools and technologies the SkyTech program uses, what each one does, what it costs and where it runs. The picture below shows the journey from finding a business to winning a customer. The architecture diagram after it shows how the same tools connect behind the scenes.

47

tools and technologies listed

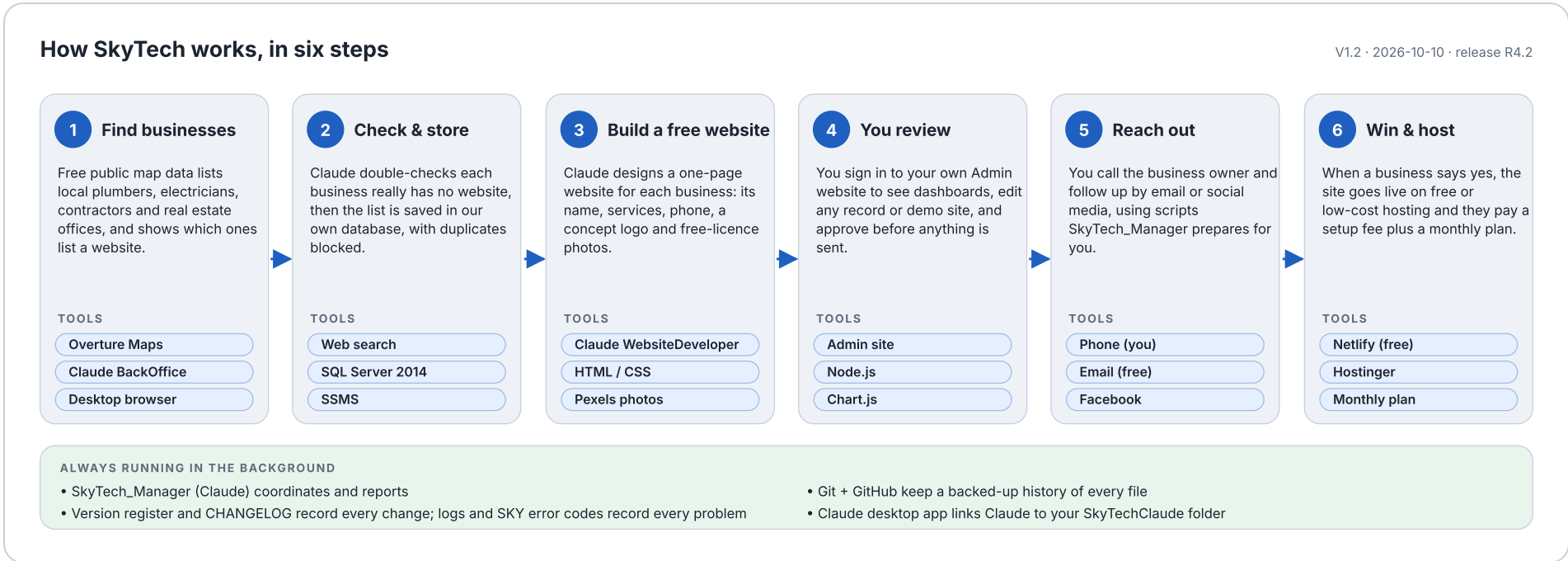
35

already in use or ready

\$0

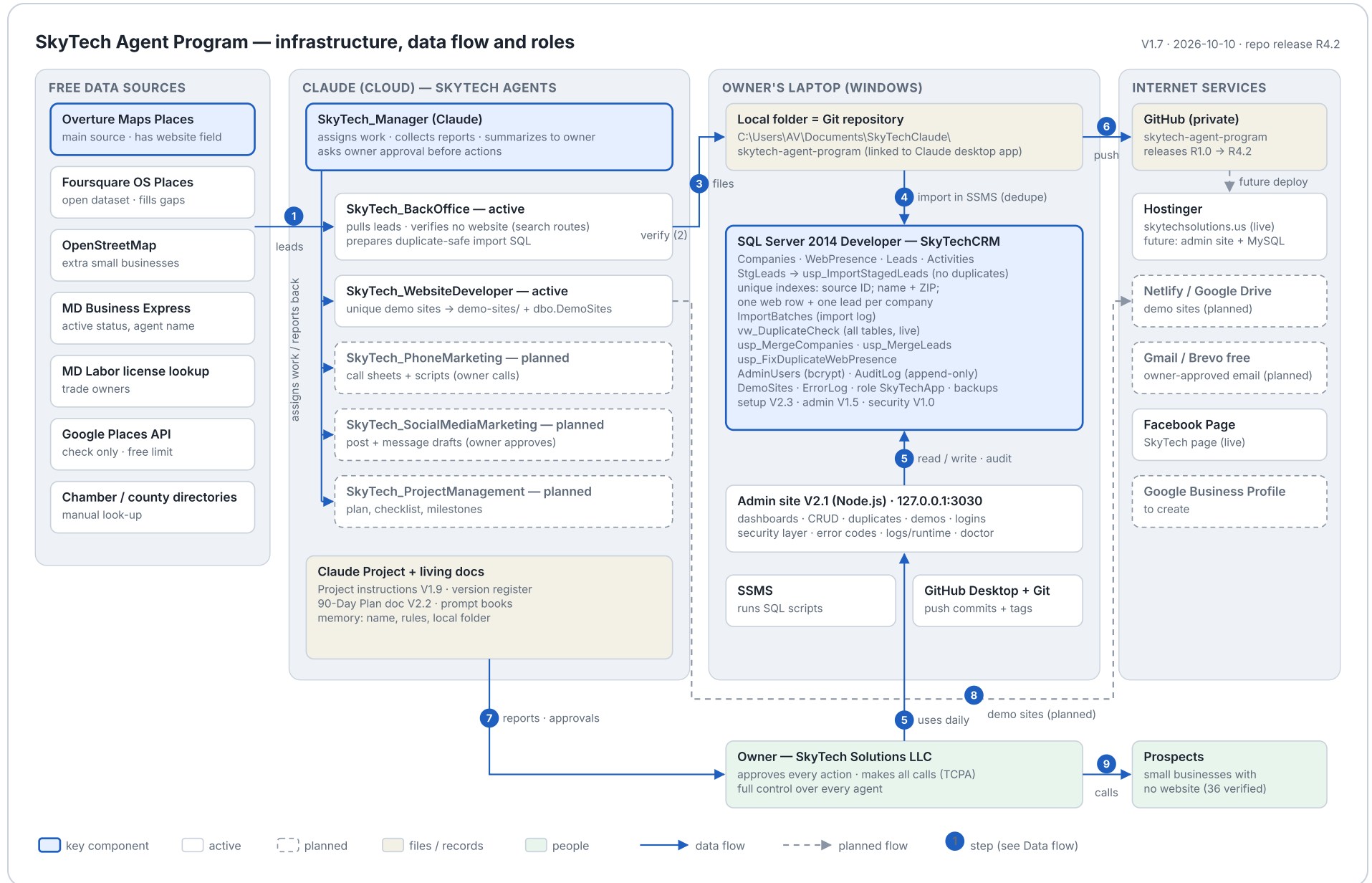
new monthly spend: 44 are free or included; the rest use things you already have: your Claude plan, the Hostinger plan for skytechsolutions.us and your phone

1. The one-minute picture



2. Architecture diagram: how the tools connect

Same tools, shown by where they live: free data sources, the Claude agents, your laptop and internet services. Numbers follow the data flow below.



Data flow, step by step

1. Lead pull — SkyTech_BackOffice collects businesses from Overture Maps Places (desktop-app browser); other free sources fill gaps.
2. Verification — BackOffice searches name, phone, address and likely domains; records a verdict and evidence for each lead.
3. Files — lead CSV, verification results and import SQL are saved in the local folder (Git repository).
4. Import — owner runs the SQL in SSMS; usp_ImportStagedLeads inserts new companies and updates existing ones (no duplicates).
5. Admin site — owner signs in, reviews dashboards, drills down, edits records, fixes any duplicates the live check finds; every change is audited.
6. Backup & history — GitHub Desktop pushes the local folder and release tags to the private GitHub repository.
7. Reporting — agents report to SkyTech_Manager; the Manager summarizes to the owner and asks approval before actions.
8. Demo sites — WebsiteDeveloper builds demos into the local folder (demo-sites/) with free-licence photos (Pexels); after owner approval they are published to free hosting (Netlify, planned) for the marketing agents.
9. Outreach — owner calls and emails prospects (approved scripts); outcomes are logged as Activities in the Admin site.
10. Errors & logs — every failure shows a SKY code and a reference; the Admin site, scripts and doctor write logs/runtime, procedures write dbo.ErrorLog; the Troubleshooting Guide gives the fix.

3. Technology stack, in plain words

1 · Finding businesses

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Overture Maps Places	A free, open database of places and businesses, published by a foundation backed by Amazon, Meta, Microsoft and TomTom.	Main list of local businesses, including whether each one lists a website.	Free (open licence)	explore.overturemaps.org, opened in the Claude desktop browser	In use
Foursquare Open Source Places	A free, open dataset of business locations.	Fills gaps in the Overture list.	Free	Downloaded file	Planned
OpenStreetMap	A world map built by volunteers, free to use.	Extra small businesses not in other lists.	Free	Internet	Planned

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Maryland Business Express	The State of Maryland's business registry search.	Confirms a business is active and shows the owner or agent name.	Free	Website (manual look-up)	Planned
MD Dept. of Labor licence look-up	The State's search for trade licences.	Confirms licences (MHIC, plumbing, electrical) before we contact a business.	Free	Website (manual look-up)	Planned
Google Places API	Google's business-listing service for programs.	Spot checks only, inside the free monthly allowance.	Free tier	Internet	Planned
Chamber and county directories	Local member and business lists.	Manual look-ups for extra leads.	Free	Websites	Planned

2 · The AI team

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Claude (Anthropic), in Cowork	An AI assistant that does the work of the named SkyTech agents.	SkyTech_Manager, BackOffice and WebsiteDeveloper: plans, checks leads, builds sites, keeps files and reports to you.	Your existing Claude plan (no added cost)	claude.ai and the Claude desktop app	In use
Claude Project	A shared space with instructions and documents that every Claude session reads.	Project instructions, version register, daily prompt files.	Included	claude.ai	In use
Claude desktop app (linked computer)	The app that lets Claude work inside your SkyTechClaude folder and use a built-in browser.	Reading and saving files, running scripts, collecting leads and photos.	Free app	Your Windows laptop	In use

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Saved skill: skytech-backoffice	A saved set of instructions Claude reuses.	Runs SkyTech_BackOffice the same way every time.	Included	Claude	In use
Scheduled tasks	Claude jobs that run on a timer.	Daily versioning run: prompt file and version register.	Included	Claude (cloud)	In use
Claude Docs living document	An online document you can read and edit.	The 90-Day Plan.	Included	claude.ai	In use

3 · Storing & checking data

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Microsoft SQL Server 2014 Developer Edition	A database: a very organised set of tables, like spreadsheets that check each other.	SkyTechCRM: companies, leads, activities and demo sites, with duplicate entries blocked.	Free edition (already installed)	Your laptop (database not created yet)	Installed
SQL Server Management Studio (SSMS)	The program used to open the database and run its scripts.	Running sql/01, the lead import, sql/04 and sql/05.	Free	Your laptop	Ready
SQL scripts (T-SQL)	Step-by-step instructions for the database.	Create the tables, import leads without duplicates, load the demo sites.	Free (ours)	sql/ folder	Ready

4 · Your Admin website

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Node.js	The engine that runs JavaScript programs on a computer.	Runs the Admin site and the demo-site builder.	Free	Your laptop	Ready

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Express, express-session, dotenv	Building blocks for a small web server.	Admin pages, sign-in sessions and settings.	Free	Your laptop	Ready
bcryptjs	A password scrambler.	Stores Admin site passwords scrambled so they cannot be read (managed on the Logins screen).	Free	Your laptop	Ready
mssql, msnodesqlv8, ODBC driver	Connectors between Node.js and SQL Server.	Lets the Admin site read and save database records.	Free	Your laptop	Ready
Security layer (security.js)	The site's locks and alarms, written by us.	Blocks password guessing, forged requests from other websites and request floods; signs you out when idle; strict browser rules (CSP).	Free (ours)	Your laptop	Ready
Logging + error codes (logger.js, errors.js)	A diary of everything the site does, and a numbered list of every possible problem.	Each problem shows a code like SKY-DB-001 that the Troubleshooting Guide explains; logs are kept 30 days with passwords hidden.	Free (ours)	Your laptop	Ready
Doctor (npm run doctor)	A one-command health check.	Checks settings, packages, database, backups and GitHub safety, and tells you exactly how to fix anything wrong.	Free (ours)	Your laptop	Ready
Chart.js	A library for drawing charts.	Dashboard charts you can click to drill down.	Free	Inside the Admin site	Ready

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
HTML, CSS, JavaScript	The basic building blocks of every web page.	The Admin screens and every demo site.	Free	Browser	In use

5 · Demo websites

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Demo-site builder (render.js, build-demos.js)	Our own template program.	Turns each business's design file or database record into a finished one-page website.	Free (ours)	admin-site folder	In use
Pexels	A free stock-photo library; photos may be used commercially.	Real photos for banners and service cards (credited in the footer).	Free licence	Internet	In use
Google Fonts	A free library of web typefaces.	Professional fonts for each design.	Free	Internet	In use
Concept logos and illustrations (SVG)	Drawings made in code, original to each site.	Concept logos and artwork until the business gives us theirs.	Free (ours)	Inside each page	In use
Netlify	Free website hosting.	A shareable link for each approved demo.	Free tier	Internet	Planned

6 · Reaching out & our presence

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Phone (owner)	You make every call yourself.	Calling business owners (US calling rules, TCPA).	Existing phone	You	Next
Gmail / Brevo free	Email sending.	Follow-up emails you approve, with an opt-out line.	Free tier	Internet	Planned

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Facebook Page	SkyTech's page on Facebook.	Posts and messages you approve.	Free	Internet	Live
Google Business Profile	Your listing on Google Maps and Search.	Being found by local customers.	Free	Internet	To create
Google Search Console	Google's website health tool.	Getting skytechsolutions.us indexed and checked.	Free	Internet	Planned
Hostinger	The web host for skytechsolutions.us.	Company website; later the Admin site.	Existing plan	Internet	Live

7 · Safety net: files & backup

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Git	A change-tracker that keeps every version of every file.	Each change is saved as a numbered release (R1.0, R2.0 ...).	Free	Your laptop	In use
GitHub (private repository)	An online, private backup of the Git history.	Off-site copy of all work: skytech-agent-program.	Free	Internet	In use
GitHub Desktop + push-to-github.bat	One-click upload tools.	Sending new releases to GitHub.	Free	Your laptop	In use
Secret scan (scan-secrets.js)	A check that runs before every upload to GitHub.	Stops passwords, keys or the .env settings file from ever being uploaded.	Free (ours)	Your laptop	Ready
SQL Server backup (usp_BackupSkyTechCRM)	A verified copy of the whole database.	Weekly backup to C:\SkyTechBackups, checked so it can really be restored.	Free	Your laptop	Ready

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Version register, INDEX, CHANGELOG	Our record books.	Which version of every file is current and what changed.	Free (ours)	config/ and repository root	In use

8 · Behind-the-scenes helpers

Tool / technology	What it is, in plain words	What we use it for	Cost	Where it runs	Status
Python 3	A popular programming language.	Lead scripts, prompt book, the diagram and this document.	Free	Your laptop (Claude runs it)	In use
ReportLab	A PDF maker for Python.	The Prompt Book PDF.	Free	Your laptop	In use
Playwright + Chromium	An automated web browser.	Turning diagrams into PDF/PNG and testing pages on phone and desktop sizes.	Free	Claude workspace	In use
pypdf	A PDF tool.	Cleans PDF properties so documents carry no AI traces.	Free	Claude workspace	In use
Markdown	A simple text format for documents.	Logs, role files and daily prompt files.	Free	Repository	In use

4. Words you may see

Word	Meaning
Database	A set of linked tables that stores records and checks them; ours is called SkyTechCRM.
CRM	Customer relationship management: keeping track of every business we contact and where each one stands.
Script	A saved list of instructions a computer follows.

Word	Meaning
Lead	A business that might become a customer.
Demo site	A free sample website we build to show a business what it could have.
Repository / release	The project folder with its full change history; a release is a numbered snapshot (for example R3.4).
Hosting	A service that keeps a website online so anyone can open it.
API	A doorway that lets one program ask another for data.
Open licence / free licence	Permission to use data or photos without paying, under stated rules.
noindex	A tag that keeps a page out of Google until we are ready to publish it.
Error code	A short label such as SKY-DB-001 that points to one problem and its fix in the Troubleshooting Guide.
Log	A file where the program writes down what it did and any problem, with the time.
CSRF / CSP	Protections that stop other websites from making your browser change your data or run unwanted code.

5. How this document is kept up to date

SkyTech_Manager updates this document whenever a tool is added, changed or retired. The tool list and the architecture diagram come from the same source files (`scripts/docs/build_tech_stack.py` and `build_infrastructure.py`), so they always match. Each update bumps the version, adds a line to the change log below, and is saved in the local folder and released to GitHub with the other artifacts.

Change log

Version	Date	Change
V1.0	2026-10-10	First version: one-minute journey picture, architecture diagram, technology stack in plain words, costs, glossary.
V1.1	2026-10-10	R4.0: security layer, logging and error codes, doctor, secret scan and verified database backups added; diagram V1.6.
V1.2	2026-10-10	R4.2: Admin site manages its own logins (create, edit, disable, reset password); diagram V1.7.

Version: V1.2 (2026-10-10) — docs/architecture/SkyTech_Technology_Stack_Use.html — V1.2. Rebuild with: python scripts/docs/build_tech_stack.py